

Newtonian mechanics practice problems

kinematics of a body on an incline

A small block slides without friction down an inclined plane starting from rest. Let s_n be the distance travelled from $t = n - 1$ to $t = n$. Then, $\frac{s_n}{s_{n+1}}$ is

1. $\frac{2n-1}{2n}$
2. $\frac{2n+1}{2n-1}$
3. $\frac{2n-1}{2n+1}$
4. $\frac{2n}{2n+1}$

A body of mass 'm' is launched up on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of friction between the body and plane is $\frac{\sqrt{x}}{5}$ if the time of ascent is half of the time of descent. The value of x is ?.

calculus application

Q1) The position vector $\mathbf{r}$ of a particle of mass m is given by

$$\mathbf{r}(t) = \alpha t^3 \hat{\mathbf{i}} + \beta t^2 \hat{\mathbf{j}}$$

where $\alpha = \frac{10}{3} \text{ ms}^{-3}$, $\beta = 5 \text{ ms}^{-2}$ and $m = 0.1 \text{ kg}$. At $t = 1 \text{ s}$, which of the following statement(s) is (are) true about the particle? (JEE adv PYQ)

1. The velocity $\mathbf{v}$ is given by $\mathbf{v} = (10\hat{\mathbf{i}} + 10\hat{\mathbf{j}}) \text{ ms}^{-1}$.
2. The angular momentum $\mathbf{L}$ with respect to the origin is $\mathbf{L} = \frac{5}{3} \hat{\mathbf{k}} \text{ Nms}$.
3. The force $\mathbf{F}$ is given by $\mathbf{F} = (\hat{\mathbf{i}} + 2\hat{\mathbf{j}}) \text{ N}$.
4. The torque $\boldsymbol{\tau}$ with respect to the origin is $\boldsymbol{\tau} = -\frac{20}{3} \hat{\mathbf{k}} \text{ Nm}$.

Q2) A particle of mass M originally at rest is subjected to a force whose direction is constant but magnitude varies with time according to the relation

$$F = F_0 \left[1 - \left(\frac{t-T}{T} \right)^2 \right] \quad (1)$$

Where F_0 and T are constants. The force acts only for the time interval 2T. The velocity v of the particle after time 2T is (JEE mains):

- (1) $2F_0T / M$
- (2) $4F_0T / 2M$
- (3) $4F_0T / 3M$
- (4) $F_0T / 3M$

Atwood system

The pulleys and strings shown in the figure 1 are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be

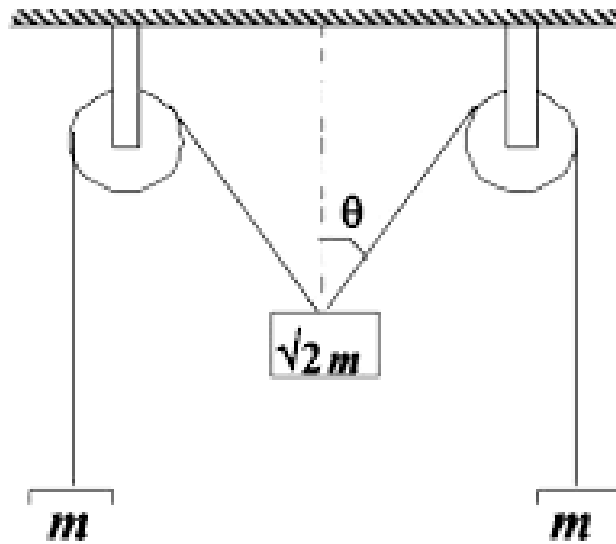


Figure 1: atwood system

- 1. 0
- 2. 30
- 3. 45
- 4. 60

Friction

Q1) A block kept on a rough inclined plane as shown in the figure 2, remains at rest up to a maximum force of 2 N down the inclined plane. The maximum external force 2 N is needed such that the block does not move up the inclined plane. The coefficient of static friction between the block and the inclined plane is

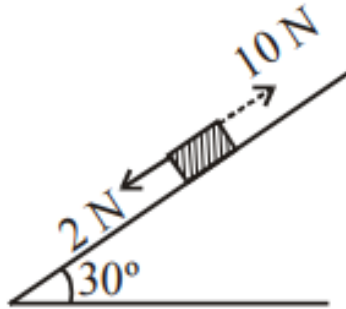


Figure 2: diagram

1. $\frac{2}{3}$
2. $\frac{\sqrt{3}}{2}$
3. $\frac{\sqrt{3}}{4}$
4. $\frac{1}{2}$

Q2)The coefficient of static friction between a wooden block of mass 0.5 kg and a vertical rough wall is 0.2. The magnitude of horizontal force that should be applied on the block to keep it adhere to the wall will be ?N. [g = 10 ms⁻²]

Q3)The coefficient of static friction between two blocks is 0.5 and the table is smooth(fig 3). The maximum horizontal force that can be applied to move the blocks together isN.

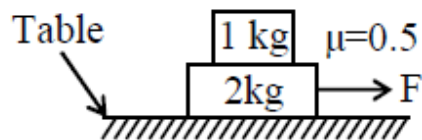


Figure 3: diagram